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SE3082 – Assignment Proposal

2 messages

it23575608 RANWALA H W <it23575608@my.sliit.lk>
To: nuwan.k@sliit.lk

Sun, Apr 19, 2026 at 7:13 PM

Dear Mr. Nuwan,

I am writing to formally submit my proposed algorithm for the SE3082 – Parallel Computing assignment.

Title of the Algorithm: Parallel Generation of the Mandelbrot Set

Problem Domain: Computer Graphics and Rendering

Brief Description:

The Mandelbrot set is a mathematical set of points in the complex plane that remain bounded under the iterative function $z_{n+1} = z_n^2 + c$, where c is a complex number. Generating an image of the set involves calculating the escape velocity for thousands of individual pixels, where the number of iterations required to exceed a specific threshold determines the pixel's color.

This algorithm demonstrates a high degree of data parallelism, as the computation for each pixel is entirely independent of all other pixels. Since there are no inter-pixel data dependencies or race conditions, the workload can be distributed seamlessly without the need for complex synchronization primitives. This inherent independence makes the algorithm an ideal candidate for demonstrating the performance scaling of parallel architectures. By partitioning the pixel grid into sub-regions, I intend to evaluate the efficiency of multiple CPU threads (OpenMP), distributed processes (MPI), and massive thread-level parallelism via GPU cores (CUDA) to achieve significant computational speedup compared to a serial implementation.

Serial C Code:

The following implementation uses a standard grid-based approach to calculate the Mandelbrot set for a specified resolution.

```
#include <stdio.h>
```

```
void generate_mandelbrot(int width, int height, int max_iter) {
```

```
    for (int y = 0; y < height; y++) {
        for (int x = 0; x < width; x++) {
            double c_re = (x - width / 2.0) * 4.0 / width;
            double c_im = (y - height / 2.0) * 4.0 / height;
            double x_curr = 0, y_curr = 0;
            int iteration = 0;
```

```
            while (x_curr * x_curr + y_curr * y_curr <= 4 && iteration < max_iter) {
```

```
                double x_new = x_curr * x_curr - y_curr * y_curr + c_re;
                y_curr = 2 * x_curr * y_curr + c_im;
                x_curr = x_new;
                iteration++;
            }
```

```
            // In a full implementation, the 'iteration' value would be mapped to a color
        }
```

```
    }
```

```
int main() {
    generate_mandelbrot(800, 600, 1000);
    return 0;
}
```

Source: Concept adapted from standard academic literature on fractal generation.

I look forward to your approval to proceed with the implementation and performance evaluation.

Best regards,

H. W. Ranwala
Year 3, Semester 1,
Bsc (Hons) in Computer Science

Nuwan Kodagoda <nuwan.k@sliit.lk>
To: RANWALA H W it23575608 <it23575608@my.sliit.lk>

Sun, Apr 19, 2026 at 10:32 PM

Classification: Public-SLIIT

Dear Student,

This is fine, please proceed.

include this email correspondence in the project submission.

Best Regards

Nuwan

From: it23575608 RANWALA H W <it23575608@my.sliit.lk>

Date: Sunday, 19/04/26 at 7:13 PM

To: Nuwan Kodagoda <nuwan.k@sliit.lk>

Subject: SE3082 – Assignment Proposal

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